

Analysis of feasibility margin as a Control Barrier Function under Input Constraints

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Abstract: Quadratic Programs (QP) subject to Control Barrier Function (CBF)-based constraints are widely employed to design safety-critical controllers. However, ensuring the feasibility of the QP under input constraints remains a significant challenge. In this work, we propose a feasibility-margin-based CBF as a proactive safety filter to guarantee the feasibility of CBF-QP with input constraints. We first characterize the feasibility margin using support functions defined by the geometry of input constraints. We then propose a novel method that employs the feasibility margin as a Control Barrier Function (FMA-CBF) for safety-critical control systems subject to polytopic input constraints. Furthermore, we formulate a unified QP that enforces both the original safety constraints and the feasibility margin constraints to guarantee feasibility. The efficacy of the proposed method is validated through double-integrator systems and unicycle robots with obstacle avoidance tasks.

Keywords: Safety-Critical Control, Control barrier function, feasibility margin, Input Constraints.

1. INTRODUCTION

In the rapidly evolving landscape of autonomous unmanned systems and robotics, ensuring safety is critical. Control Barrier Functions (CBFs) have emerged as a systematic and constructive mathematical framework for synthesizing safety-critical controllers (Ames et al. (2019)). By encoding safety requirements as forward invariant sets within the state space, CBFs allow for the rigorous construction of optimization-based controllers, typically formulated as Quadratic Programs (QPs), which minimally deviate from a nominal performance objective while strictly enforcing safety constraints. This methodology has seen successful applications across diverse domains, ranging from adaptive cruise control (Ames et al. (2014)) to legged locomotion (Nguyen and Sreenath (2015)) and multi-agent coordination (Borrmann et al. (2015)).

However, a pervasive challenge in the practical implementation of CBF-based controllers is the presence of physical actuation limits, or input constraints. The theoretical guarantees of standard CBFs often rely on the assumption that the set of admissible control inputs is sufficient to satisfy the barrier condition $\dot{h}(\mathbf{x}, u) \geq -\alpha(h(\mathbf{x}))$ at all times. In reality, actuators (e.g., motors, thrusters, brakes) are subject to hard saturation limits, modeled as a bounded polytope $\mathcal{U} \subset \mathbb{R}^m$. When the system approaches the boundary of the safe set with high velocity, the control effort required to maintain invariance may exceed the physical capability of the actuators.

This phenomenon leads to the infeasibility of the QP, causing the system to enter an *Inevitable Collision State* (ICS) (Fraichard and Asama (2003)) even before the safety constraint is explicitly violated. Consequently, bridging the gap between theoretical safety guarantees and phys-

ical input constraints remains a non-trivial and critical bottleneck in modern control theory.

Addressing the challenge of feasibility for CBF-QPs under input constraints has been a focal point of recent research. Existing strategies can be broadly categorized into three main approaches, each with distinct limitations:

First, *Backup Control Barrier Functions* ensure safety by verifying the existence of a feasible trajectory returning to a safe set (Gurriet et al. (2018)). While this approach provides a rigorous safety guarantee, its primary drawback lies in the dependency on a specific braking maneuver (Singletary et al. (2022); van Wijk et al. (2025)). Designing such backup policies is often heuristic, conservative, and difficult to generalize for complex nonlinear systems. Second, *MPC-CBF* approaches incorporate finite-time prediction horizons into the barrier function formulation to anticipate future constraint violations (Zeng et al. (2021)). However, this comes at the cost of high computational burden associated with Model Predictive Control. Third, *Hamilton-Jacobi(HJ) Reachability* analysis offers a theoretically complete solution by computing the exact viability kernel (or backward reachable set) (Hsu et al. (2023)). This method defines the maximal set of states from which safety can be maintained under input limits. Despite its theoretical rigor, computing the solution to the associated Hamilton-Jacobi-Isaacs partial differential equations is computationally prohibitive for high-dimensional systems.

A more direct approach is to analyze the intersection between the control input polytope and the half-space defined by the CBF constraint (Parwana et al. (2025)). The core difficulty lies in the geometric complexity: the condition for feasibility is that the intersection of the

admissible control set and the safe control set must be non-empty.

This paper proposes a novel differentiable safety framework for input-constrained affine systems. The main contributions are as follows:

- (1) *Feasibility margin characterization*: We rigorously characterize the feasibility margin via support functions and provide an explicit analysis of the non-smoothness arising from the geometry of polytopic input constraints.
- (2) *Smooth approximation with analytical derivatives*: To overcome the non-differentiability challenge, we develop a smooth approximation of the feasibility margin and derive its closed-form derivatives with respect to the system dynamics, thereby enabling efficient gradient-based optimization.
- (3) *Unified control framework and validation*: We formulate a unified high-order time-varying CBF-QP controller that integrates the smoothed feasibility margin to ensure constraint satisfaction. Numerical simulations demonstrate that the proposed framework maintains safety under tightly constrained scenarios where conventional methods suffer from infeasibility.

2. PRELIMINARIES

2.1 Control Barrier Functions

Consider a nonlinear control-affine system with state $\mathbf{x} \in \mathbb{R}^n$ and control input $\mathbf{u} \in \mathbb{R}^m$, described by the dynamics

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u}, \quad (1)$$

where $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are locally Lipschitz continuous vector fields. Our objective is to design a locally Lipschitz continuous controller $\mathbf{k} : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^m$, denoted as $\mathbf{u} = \mathbf{k}(\mathbf{x}, t)$, such that the closed-loop system:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{k}(\mathbf{x}, t) \quad (2)$$

guarantees safe behavior. For any initial condition $\mathbf{x}(0) = \mathbf{x}_0 \in \mathbb{R}^n$, a unique solution $\mathbf{x}(t)$ to (2) exists for all $t \geq 0$.

We define safety as the requirement that the system state $\mathbf{x}(t)$ remains within a specified *safe set* $\mathcal{C} \subset \mathbb{R}^n \times \mathbb{R}$ for all time.

Definition 1. *The system (1) is safe with respect to the set $\mathcal{C}$ if for every initial condition satisfying $(\mathbf{x}_0, 0) \in \mathcal{C}$, the solution satisfies $(\mathbf{x}(t), t) \in \mathcal{C}$ for all $t \geq 0$. This property implies that the set $\mathcal{C}$ is forward invariant along the trajectories of (1).*

Let the safe set $\mathcal{C}$ be defined as the zero-superlevel set of a continuously differentiable function $h : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$

$$\begin{aligned} \mathcal{C} &= \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : h(\mathbf{x}, t) \geq 0\}, \\ \partial\mathcal{C} &= \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : h(\mathbf{x}, t) = 0\}. \end{aligned} \quad (3)$$

We assume that the boundary $\partial\mathcal{C}$ is nonempty and has no isolated points, and that 0 is a regular value of h (i.e., $\nabla_{\mathbf{x}}h(\mathbf{x}, t) \neq \mathbf{0}$ for all $(\mathbf{x}, t) \in \partial\mathcal{C}$).

Control Barrier Functions (CBFs) provide a constructive method to synthesize controllers with formal safety guarantees. The time evolution of h along the system dynamics is given by

$$\dot{h}(\mathbf{x}, t, \mathbf{u}) = \frac{\partial h}{\partial t} + L_{\mathbf{f}}h(\mathbf{x}, t) + L_{\mathbf{g}}h(\mathbf{x}, t)\mathbf{u}, \quad (4)$$

where $L_{\mathbf{f}}h(\mathbf{x}, t) \triangleq \frac{\partial h}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x})$ and $L_{\mathbf{g}}h(\mathbf{x}, t) \triangleq \frac{\partial h}{\partial \mathbf{x}}\mathbf{g}(\mathbf{x})$ denote the Lie derivatives along $\mathbf{f}$ and $\mathbf{g}$, respectively.

Definition 2. (Ames et al. (2019)) *A continuously differentiable function h is a Control Barrier Function (CBF) for the system (2) on $\mathcal{C}$ if there exists an extended class $\mathcal{K}_{\infty}$ function α such that for all $(\mathbf{x}, t) \in \mathcal{C}$*

$$\sup_{\mathbf{u} \in \mathbb{R}^m} [\dot{h}(\mathbf{x}, t, \mathbf{u})] \geq -\alpha(h(\mathbf{x}, t)). \quad (5)$$

Given a CBF, the following equation establishes the safety guarantee for the control input set

$$\mathcal{U}_{\text{cbf}} = \{\mathbf{u} \in \mathbb{R}^m : [\dot{h}(\mathbf{x}, t, \mathbf{u})] \geq -\alpha(h(\mathbf{x}, t))\}. \quad (6)$$

If the system has a high relative degree, specifically if $L_{\mathbf{g}}h(\mathbf{x}, t) \equiv \mathbf{0}$, the control input $\mathbf{u}$ does not appear explicitly in (4), rendering (5) unenforceable directly. In such cases, techniques such as High-Order CBFs (HOCBFs) (Xiao and Belta (2022)) are employed. The core idea of HOCBFs is to define an extended barrier function h_e that incorporates the desired decay of h

$$h_e(\mathbf{x}, t) = \dot{h}(\mathbf{x}, t) + \alpha(h(\mathbf{x}, t)). \quad (7)$$

The maintenance of $h_e(\mathbf{x}, t) \geq 0$ ensures the satisfaction of the safety condition of h . Let $\mathcal{C}_e$ denote the zero-superlevel set of h_e .

While (5) guarantees the existence of a safe control in the unconstrained case (provided $L_{\mathbf{g}}h \neq \mathbf{0}$), physical system limits on the control $\mathbf{u} \in \mathcal{U}$ may render the safety condition infeasible. Specifically, at certain states $\mathbf{x} \in \mathcal{C}$, it may be impossible to satisfy the decay requirement even with maximal control effort

$$\sup_{\mathbf{u} \in \mathcal{U}} [L_{\mathbf{g}}h(\mathbf{x}, t)\mathbf{u}] < -\alpha(h(\mathbf{x}, t)) - L_{\mathbf{f}}h(\mathbf{x}, t) - \frac{\partial h}{\partial t}. \quad (8)$$

States where this condition holds constitute an “infeasible region”, from which safety cannot be maintained. To prevent the system from entering such regions, a *feasibility margin* must be introduced.

2.2 feasibility margin

The admissible control set $\mathcal{U}$ is modeled as a convex polytope, constituted by the convex hull of N vertices $\{v_1, v_2, \dots, v_N\}$ in an m -dimensional space

$$\mathcal{U} = \text{Conv}\{v_1, v_2, \dots, v_N\} \subset \mathbb{R}^m \quad (9)$$

This implies that any valid control input $\mathbf{u} \in \mathcal{U}$ can be expressed as a convex combination of these N vertices:

$$\mathbf{u} = \sum_{i=1}^N \lambda_i v_i, \quad \text{s.t.} \quad \sum_{i=1}^N \lambda_i = 1, \lambda_i \geq 0$$

To guarantee the feasibility of the CBF-based Quadratic Program (QP) subject to input constraints, it is imperative to ensure a non-empty intersection between the admissible control set $\mathcal{U}$ and the safe control set $\mathcal{U}_{\text{cbf}}$. To this end, we introduce the concept of the *feasible CBF space*.

Definition 3. (Parwana et al. (2025)) *At a state $\mathbf{x} \in \mathcal{X}$, the feasible CBF space $\mathcal{U}_c \subset \mathbb{R}^m$ is the intersection of set $\mathcal{U}_{\text{cbf}}(\mathbf{x}, t)$ with the input domain $\mathcal{U}$*

$$\mathcal{U}_c(\mathbf{x}, t) = \mathcal{U}_{\text{cbf}}(\mathbf{x}, t) \cap \mathcal{U}. \quad (10)$$

The feasibility margin quantifies whether the intersection between the admissible control set $\mathcal{U}$ and the safe control set $\mathcal{U}_{\text{cbf}}$ is nonempty (see Figure 1)

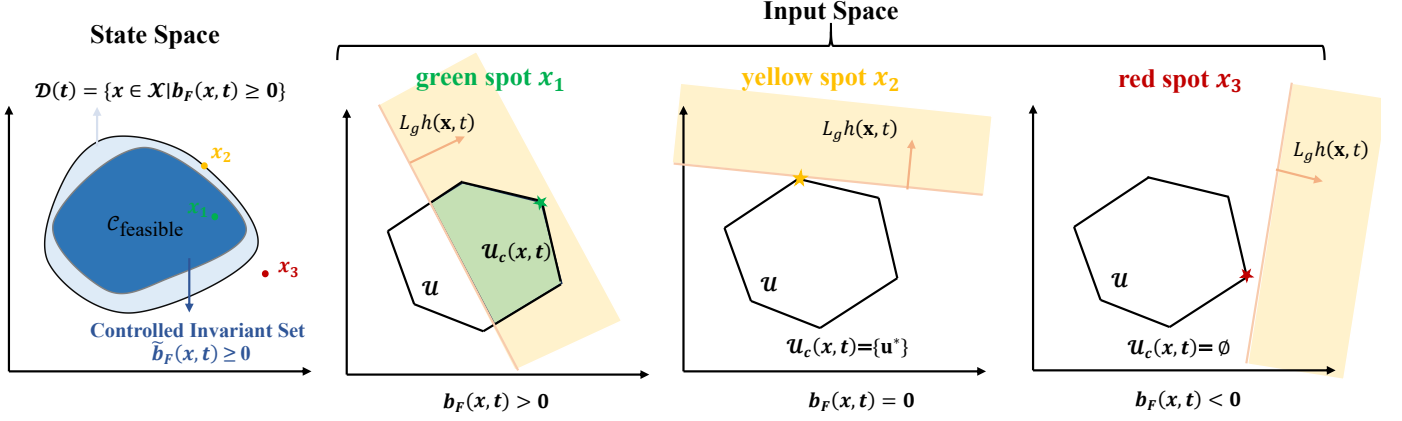


Fig. 1. **Geometric interpretation of the feasibility margin $b_F(\mathbf{x}, t)$ in the Input Space.** The feasibility margin quantifies whether the intersection between the admissible control polytope $\mathcal{U}$ and the safe control set $\mathcal{U}_{\text{cbf}}$ is non-empty. The star (*) corresponds to the specific vertex $u^* \in \mathcal{U}$ that maximizes the support function along the safety gradient $L_g h(\mathbf{x}, t)$, thereby determining the value of the margin. Consequently, $b_F(\mathbf{x}, t) \geq 0$ (Green/Yellow) indicates that the feasible control space $\mathcal{U}_c$ is non-empty, while $b_F(\mathbf{x}, t) < 0$ (Red) signifies that the intersection is empty, leading to infeasibility.

As discussed in Section 3, the smooth lower approximation $\tilde{b}_F(\mathbf{x}, t)$ is designed as a Control Barrier Function (FMA-CBF) to explicitly maintain $b_F \geq 0$, thereby guaranteeing the feasibility of the optimization problem

Definition 4. Define the feasibility margin $b_F(\mathbf{x}, t)$ as the difference between the maximum available control authority and the minimum authority required to maintain safety

$$b_F(\mathbf{x}, t) = \frac{\partial h}{\partial t} + L_f h(\mathbf{x}, t) + \alpha(h(\mathbf{x}, t)) + \sup_{u \in \mathcal{U}} (L_g h(\mathbf{x}, t)u) \quad (11)$$

3. ANALYSIS OF FEASIBILITY MARGIN AS CONTROL BARRIER FUNCTION

To guarantee the solvability of the CBF-QP subject to input constraints, it is essential that the feasible CBF space remains non-empty. Leveraging the concept of the feasibility margin derived previously, we translate this geometric condition into a scalar quantification: the feasibility of the CBF-QP is ensured if and only if the feasibility margin is non-negative.

3.1 Analysis of feasibility margin

Definition 5 (Support Function). Given a non-empty compact convex set $\mathcal{U} \subset \mathbb{R}^m$, the support function $\sigma_{\mathcal{U}} : \mathbb{R}^m \rightarrow \mathbb{R}$ is defined as the supremum of the projection of the set onto a given direction vector $p \in \mathbb{R}^m$

$$\sigma_{\mathcal{U}}(p) \triangleq \sup_{u \in \mathcal{U}} p^\top u. \quad (12)$$

Geometrically, $\sigma_{\mathcal{U}}(p)$ describes the signed distance from the origin to the supporting hyperplane of $\mathcal{U}$ with normal vector p (Schneider (2013)).

The term $\sup_{u \in \mathcal{U}} (L_g h(\mathbf{x}, t)u)$ is the Support Function of the set $\mathcal{U}$ in the direction $p = (L_g h(\mathbf{x}, t))^T$. For a polytope $\mathcal{U} = \text{Conv}\{v_1, \dots, v_N\}$, the value of the support function equals the maximum value of the projection onto all vertices in that direction

$$\sigma_{\mathcal{U}}(L_g h(\mathbf{x}, t)) = \max_{i \in \{1, \dots, N\}} (L_g h(\mathbf{x}, t)v_i) \quad (13)$$

Therefore, the explicit expression for the non-smooth feasibility margin is

$$b_F(\mathbf{x}, t) = \frac{\partial h}{\partial t} + L_f h(\mathbf{x}, t) + \alpha(h(\mathbf{x}, t)) + \sigma_{\mathcal{U}}(L_g h(\mathbf{x}, t)) \quad (14)$$

Note that $b_F(\mathbf{x}, t) < 0$ when $\mathcal{U}_c(\mathbf{x}, t) = \emptyset$. The feasible state space is defined as the set of state $D_0(t) = \{x \in \mathcal{X} \mid b_F(\mathbf{x}, t) \geq 0\}$. We then aim to find a CBF $\tilde{b}_F(\mathbf{x}, t)$, with class- $\mathcal{K}$ function $\alpha_b(\tilde{b}_F(\mathbf{x}, t))$, that renders $D_0(t)$ forward invariant by imposing the following condition

$$\dot{\tilde{b}}_F(\mathbf{x}, t, u) \geq -\alpha_b(\tilde{b}_F(\mathbf{x}, t)) \quad (15)$$

Remark 1. The max operator in (13) causes $\nabla b_F(\mathbf{x}, t)$ to have jump discontinuities when the active vertex switches, rendering b_F unsuitable as a CBF directly due to undefined Lie derivatives and control chattering.

3.2 Feasible Margin Control Barrier Function

To enable gradient-based control, we propose a smooth lower approximation of the feasible margin $\tilde{b}_F(\mathbf{x}, t)$ as FMA-CBF in this section.

Let $\eta > 0$ be the smoothing parameter. The smooth feasibility margin control barrier function $\tilde{b}_F(\mathbf{x}, t)$ is defined as

$$\tilde{b}_F(\mathbf{x}, t) = \underbrace{\frac{\partial h}{\partial t}}_{\text{Term 1}} + \underbrace{\nabla_x h^T f}_{\text{Term 2}} + \underbrace{\alpha(h)}_{\text{Term 3}} + \underbrace{\frac{1}{\eta} \ln \left(\sum_{i=1}^N \exp(\eta \nabla_x h^T g v_i) \right) - \frac{\ln(N)}{\eta}}_{\text{Term 4}} \quad (16)$$

Remark 2. The approximation satisfies the bounds:

$$\max_i(z_i) \leq \frac{1}{\eta} \ln \left(\sum \exp(\eta z_i) \right) \leq \max_i(z_i) + \frac{\ln(N)}{\eta}$$

We now present the following theorem that states the sufficiency of FMA-CBF condition (15) for feasibility of CBF-QP under input constraints.

Theorem 1. *Let $b_F(\mathbf{x}, t)$ be the feasibility margin of the feasible CBF space $\mathcal{U}_c(\mathbf{x}, t)$ in the control domain defined in (11). For $0 < \epsilon \ll 1$, let $\mathcal{D}_\epsilon(t) = \{\mathbf{x} \in \mathcal{X} \mid b_F(\mathbf{x}, t) \geq \epsilon\}$ be the feasible state space. Then the CBF-QP is persistently feasible if there exists a set $\mathcal{C}_{\text{feasible}} \subset \mathcal{D}_\epsilon$ and a function $\tilde{b}_F(\mathbf{x}, t) : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\tilde{b}_F(\mathbf{x}, t)$ is a control barrier function on $\mathcal{C}_{\text{feasible}}$ and $x(0) \in \mathcal{C}_{\text{feasible}}$.*

Proof. *Step 1: Forward Invariance of $\mathcal{C}_{\text{feasible}}$.* By hypothesis, $\tilde{b}_F(\mathbf{x}, t)$ is a CBF on $\mathcal{C}_{\text{feasible}}$, so there exists a class- $\mathcal{K}$ function α_b and a controller u satisfying $\dot{\tilde{b}}_F(\mathbf{x}, t, u) \geq -\alpha_b(\tilde{b}_F(\mathbf{x}, t))$. Given $x(0) \in \mathcal{C}_{\text{feasible}}$, forward invariance of CBFs guarantees $\mathbf{x}(t) \in \mathcal{C}_{\text{feasible}}$ and hence $\tilde{b}_F(\mathbf{x}(t), t) \geq 0$ for all $t \geq 0$.

Step 2: Persistent Feasibility. Since $\mathcal{C}_{\text{feasible}} \subset \mathcal{D}_\epsilon(t) = \{\mathbf{x} \mid b_F(\mathbf{x}, t) \geq \epsilon\}$, we have $b_F(\mathbf{x}(t), t) \geq \epsilon > 0$ for all $t \geq 0$. By Definition 4, $b_F(\mathbf{x}, t) > 0$ implies $\sup_{u \in \mathcal{U}} [\frac{\partial h}{\partial t} + L_f h + L_g h u + \alpha(h)] > 0$, so the set of controls satisfying the CBF condition is non-empty. Thus the QP always admits a solution and the closed-loop system is persistently feasible.

The differentiability of the Log-Sum-Exp approximation allows us to compute the gradient analytically, which is essential for the QP formulation. The detailed derivation is omitted here for brevity.

The total gradient of the smoothed feasible margin $\tilde{b}_F(\mathbf{x}, t)$ is given by

$$\begin{aligned} \nabla_x \tilde{b}_F &= \frac{\partial}{\partial t} (\nabla_x h) + H_x^x f + J_f^T \nabla_x h + \alpha' \nabla_x h \\ &+ \sum_{i=1}^N w_i (H_{g v_i}^x + J_{g v_i}^T \nabla_x h) \end{aligned} \quad (17)$$

where $H_x^h = \frac{\partial^2 h}{\partial \mathbf{x}^2}$ denotes the Hessian of h with respect to $\mathbf{x}$, $J_f = \frac{\partial f}{\partial \mathbf{x}}$ and $J_{g v_i} = \frac{\partial (g v_i)}{\partial \mathbf{x}}$ are the Jacobians of f and $g v_i$ respectively, and $w_i(\mathbf{x}, t)$ represents the softmax weights arising from the derivative of the LSE function, satisfying $\sum_{i=1}^N w_i = 1$.

Based on the gradient derived in (17), we define the Lie derivatives of the new barrier function $\tilde{b}_F$ with respect to the system dynamics(1).

The Lie derivative along the drift vector fields $f(\mathbf{x})$, denoted as $L_f \tilde{b}_F$

$$L_f \tilde{b}_F(\mathbf{x}, t) = \nabla \tilde{b}_F(\mathbf{x}, t)^T f(\mathbf{x}) \quad (18)$$

Similarly, the Lie derivative along the control vector fields $g(\mathbf{x})$, denoted as $L_g \tilde{b}_F$

$$L_g \tilde{b}_F(\mathbf{x}, t) = \nabla \tilde{b}_F(\mathbf{x}, t)^T g(\mathbf{x}) \quad (19)$$

3.3 Unified QP Optimization Formulation

To guarantee safety and feasibility, the controller must ensure two conditions simultaneously: satisfying the original safety constraint (i.e., $h(\mathbf{x}, t) \geq 0$) and maintaining the capability to remain safe in the future (i.e., $\tilde{b}_F(\mathbf{x}, t) \geq 0$).

First, we impose the original CBF condition to prevent immediate violation of the safety set $\mathcal{C}$

$$\frac{\partial h}{\partial t} + L_f h(\mathbf{x}, t) + L_g h(\mathbf{x}, t) u \geq -\alpha(h(\mathbf{x}, t)) \quad (20)$$

Second, to prevent the system from entering an infeasible state (where input constraints would make safety impossible), we impose the feasibility margin CBF condition

$$\dot{\tilde{b}}_F(\mathbf{x}, t, u) \geq -\gamma_F(\tilde{b}_F(\mathbf{x}, t)) \quad (21)$$

where γ_F is a class $\mathcal{K}$ function tuning the decay rate of the feasibility margin.

The resulting Optimization-Based Controller (QP) minimizes the deviation from the nominal controller u_{nom} while simultaneously satisfying both barrier constraints and physical input constraints

$$\begin{aligned} (u^*, \delta^*) &= \underset{u \in \mathbb{R}^m, \delta \in \mathbb{R}}{\text{argmin}} \quad \frac{1}{2} \|u - u_{\text{nom}}\|^2 + \frac{1}{2} p \delta^2 \\ \text{s.t.} \quad &\frac{\partial h}{\partial t} + L_f h(\mathbf{x}, t) + L_g h(\mathbf{x}, t) u \geq -\alpha(h(\mathbf{x}, t)) \\ &L_f \tilde{b}_F(\mathbf{x}, t) + L_g \tilde{b}_F(\mathbf{x}, t) u \geq -\gamma_F(\tilde{b}_F(\mathbf{x}, t)) + \delta \\ &u \in \mathcal{U} \end{aligned} \quad (22)$$

This formulation ensures that the control input u is safe, feasible, and admissible with respect to the actuator limits.

Remark 3. *The FMA Condition ($L_f \tilde{b}_F + L_g \tilde{b}_F u \geq -\gamma_F(\tilde{b}_F) + \delta$), serves as a proactive safety filter. Unlike the standard CBF condition (which prevents immediate safety violations via $\frac{\partial h}{\partial t} + L_f h + L_g h u \geq -\alpha(h)$), the FMA condition regulates the rate at which the “feasibility margin” decays. Physically, this ensures that the system maintains sufficient control authority (braking distance or maneuvering power) within the actuator limits $\mathcal{U}$.*

4. SIMULATION RESULTS

We present two case studies comparing the proposed FMA-CBF against a standard HOCBF baseline.

4.1 Case Study 1: Double Integrator with Input Constraints

A vehicle (double integrator) tracks a desired velocity $v_{\text{des}} = 25$ m/s while avoiding a static obstacle at $p_{\text{obs}} = 100$ m, starting from $x_0 = [0, 5]^T$. The safety constraint is $h(\mathbf{x}) = p_{\text{obs}} - p - D_{\text{safe}} \geq 0$, and the input set is

$$\mathcal{U} = \{u \in \mathbb{R} \mid u_{\min} \leq u \leq u_{\max}\}, \quad (23)$$

with $u_{\max} = 5.0$ m/s², $u_{\min} = -5.0$ m/s². The HOCBF gains are $\alpha_1 = \alpha_2 = 1.0$, and the FMA-CBF uses $\eta = 100$, $\gamma_F = 0.5$.

Two controllers are compared: (1) *Standard HOCBF*, which enforces $\ddot{h} + \alpha_1(\dot{h}) + \alpha_2(h) \geq 0$ without awareness of future infeasibility; (2) *FMA-CBF* (proposed), which incorporates $\tilde{b}_F(\mathbf{x})$ as an additional barrier constraint to proactively regulate control authority.

As shown in Fig. 2, the Standard HOCBF fails to anticipate the required braking force exceeding $u_{\min}$: the QP becomes infeasible at $t \approx 4$ s and the vehicle collides

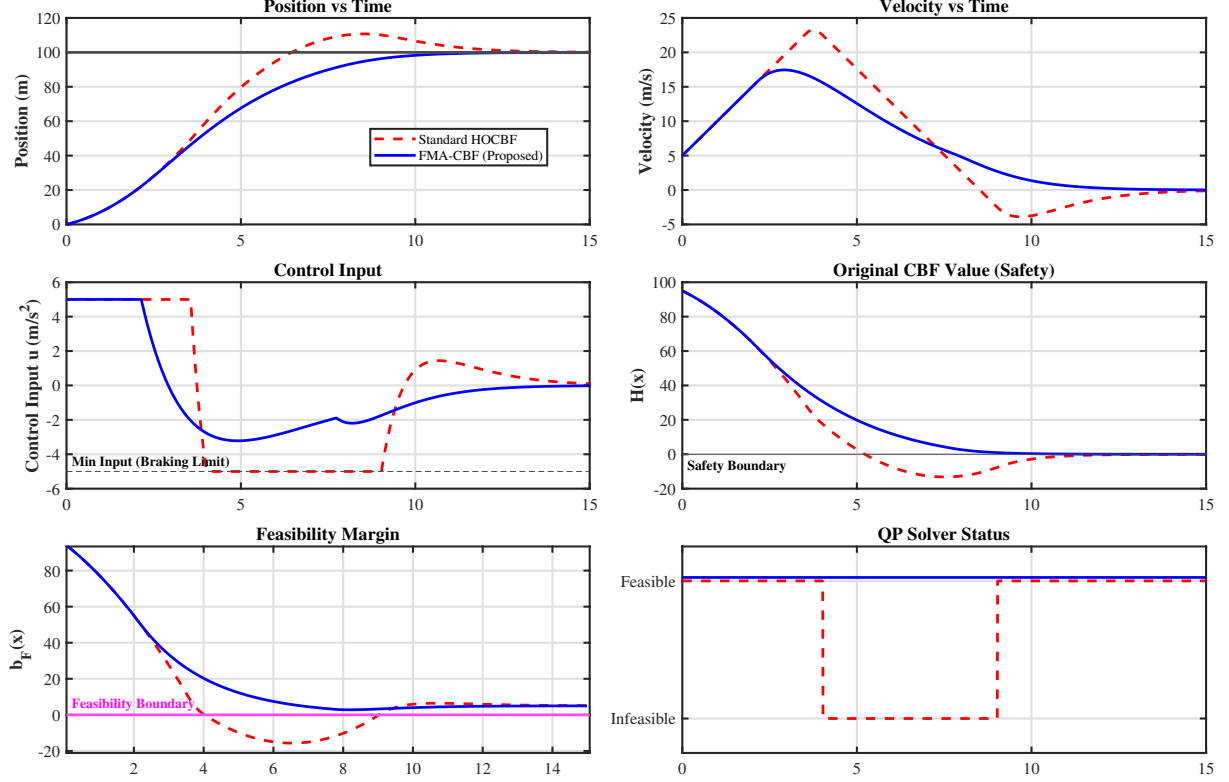


Fig. 2. Case 1 results (red dashed: Standard HOCBF; blue solid: FMA-CBF): Position, Velocity, Control Input, CBF Value, feasibility margin $b_F(\mathbf{x})$, and QP Solver Status.

with the obstacle. In contrast, the FMA-CBF uses the feasibility margin $b_F(\mathbf{x})$ as an early warning signal; once $b_F(\mathbf{x})$ approaches zero, the controller initiates braking at $t \approx 3$ s, keeping the system within the recursively feasible set $\mathcal{D}$ and maintaining safety throughout.

4.2 Case Study 2: Unicycle Obstacle Avoidance

A unicycle robot navigates from $[0, 0]^T$ to $[4, 4]^T$ while avoiding a dynamic obstacle (radius $r_o = 0.5$ m) on an intercept course, maintaining safe distance $r_{\text{safe}} = r_r + r_o + 0.1 = 0.85$ m ($r_r = 0.25$ m). The kinematic model is

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}, \quad (24)$$

with polytopic input constraints $\mathcal{U} = \{[v, \omega]^T \mid 0 \leq v \leq 2.0 \text{ m/s}, |\omega| \leq 1.5 \text{ rad/s}\}$, $\alpha_{cbf} = 2.0$, $\eta = 10$, $\gamma_F = 5.0$, slack penalty $p = 10^6$.

The baseline uses the *Center-Point Transformation* (Huang et al. (2023)): a reference point $p_c = [p_x + l \cos \theta, p_y + l \sin \theta]^T$ ($l = 0.15$ m) reduces the relative degree to 1, with barrier function $h(\mathbf{x}, t) = \|p_c - p_{\text{obs}}(t)\|^2 - r_{\text{safe}}^2$. This approach does not account for future loss of feasibility under input constraints. The proposed FMA-CBF augments the baseline with the feasibility margin constraint $\dot{b}_F(\mathbf{x}, t, u) \geq -\gamma_F(b_F(\mathbf{x}, t))$.

As shown in Fig. 3, the FMA-CBF initiates the turning maneuver earlier than the Standard CBF, maintaining $b_F > 0$ throughout. The Standard CBF delays the turn

until the feasibility margin drops below zero, causing the QP to become infeasible and ultimately leading to a collision.

5. CONCLUSION

This paper establishes a differentiable safety framework to guarantee the feasibility of Control Barrier Functions (CBFs) under input constraints. Specifically, the feasibility margin is defined to rigorously quantify the intersection between the admissible control polytope and the safe control set. The Feasibility Margin CBF (FMA-CBF) is proposed as a proactive safety filter to enforce this margin constraint, employing a smooth approximation for differentiability. Theoretical analysis confirms that the invariance of the feasibility margin guarantees persistent feasibility. Finally, simulations on double-integrator and unicycle systems demonstrate the method's capability to maintain safety in tightly constrained environments.

Future work will explore robustness extensions to handle model uncertainty and external disturbances. Another promising direction is generalizing the framework to systems subject to multiple CBF constraints, where the interaction among competing safety requirements may further complicate the feasibility analysis.

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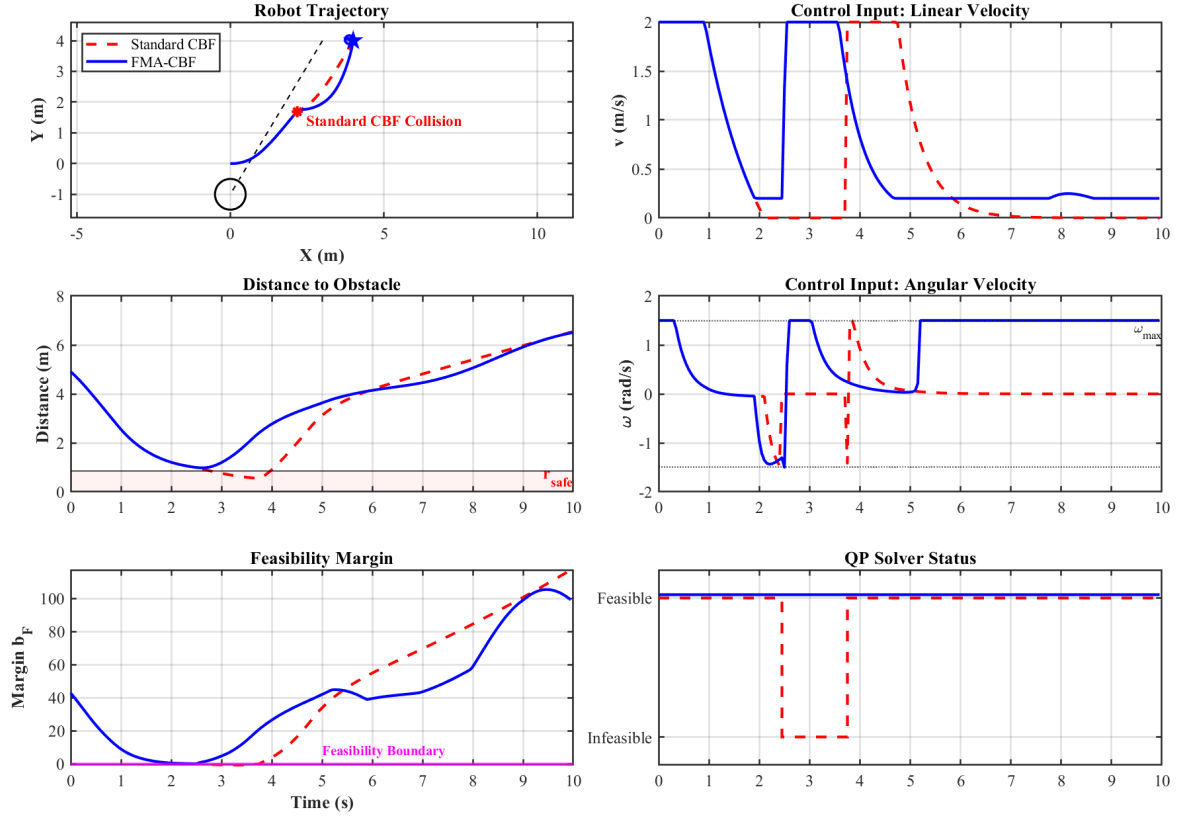


Fig. 3. Case 2 results (red dashed: Standard HOCBF; blue solid: FMA-CBF): Robot Trajectory, Linear/Angular Velocity, Distance to Obstacle, Feasibility Margin, and QP Solver Status.

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